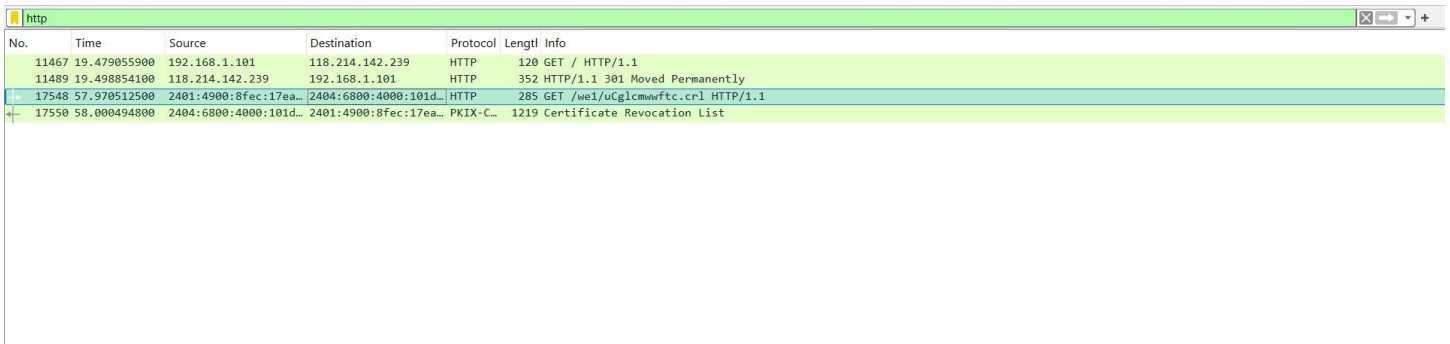


DAY-1

Day 1: Introduction to Network Protocols Theory (1 hour): Overview of DNS, DHCP, HTTP/HTTPS, FTP, and SNMP. **Practical (1 hour):** Observe DNS and HTTP traffic using Wireshark. **Task:** Write a report on the role of DNS and HTTP in web communication.

[Q-1] Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

- Capture DNS and HTTP/HTTPS packets using Wireshark while opening www.spotify.com



The screenshot shows a Wireshark packet capture window with the filter 'http'. The packet list contains five entries:

No.	Time	Source	Destination	Protocol	Length	Info
11467	19.479055900	192.168.1.101	118.214.142.239	HTTP	120	GET / HTTP/1.1
11489	19.498854100	118.214.142.239	192.168.1.101	HTTP	352	HTTP/1.1 301 Moved Permanently
17548	57.970512500	2401:4900:8fec:17ea::	2404:6800:4000:101d::	HTTP	285	GET /we1/uGlcmmwftc.crl HTTP/1.1
17550	58.000494800	2404:6800:4000:101d::	2401:4900:8fec:17ea::	PKIX-C...	1219	Certificate Revocation List

- **Explanation**

- The DNS query asks the DNS server to find the IP address of www.spotify.com.
- The DNS response returns the IP address of Spotify's server.
- After obtaining the IP address, the browser establishes a secure HTTPS connection.
- HTTPS packets begin with the TLS handshake to encrypt communication.

[Q-2] Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

○ Command

- nslookup www.flipkart.com

```
C:\Users\HET>nslookup www.flipkart.com
Server: UnKnown
Address: 2401:4900:50:9::209

Non-authoritative answer:
Name: flipkart.com
Address: 163.53.76.86
Aliases: www.flipkart.com
```

○ Explanation

- The DNS server translated the domain name www.flipkart.com into its IP address.

○ This means:

- Humans remember website names.
- Computers communicate using IP addresses.
- DNS acts like the Internet's phonebook by converting names into IP addresses.

[Q-3] Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

HTTP	HTTPS
Data is not encrypted.	Data is encrypted using SSL/TLS.
Less secure.	Highly secure.
Uses Port 80.	Uses Port 443.

○ Real-world Example

- I use **Spotify**, **Google**, and **Flipkart**, which all use HTTPS to securely transfer login information, passwords, and personal data.

[Q-4] Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe. Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.

○ Filter Used

- Dns

○ Website Searched

- www.canva.com

○ DNS Packet Sequence

1. Standard Query for **A Record** of www.canva.com.
2. Standard Query for **AAAA Record** (IPv6).
3. DNS Server sends Standard Query Response with IPv4 address.
4. DNS Server sends Standard Query Response with IPv6 address (if available).
5. Browser connects to the returned IP address.

No.	Time	Source	Destination	Protocol	Length	Info
17896	52.875710200	192.168.1.101	192.168.1.1	DNS	87	Standard query 0x77bd A media.canva.com
18069	52.990902800	192.168.1.1	192.168.1.101	DNS	169	Standard query response 0x77bd A media.canva.com CNAME media.canva.com.cdn.cloudflare.net A 104.16.103.112 A 104.16.102.112
18196	52.993622000	192.168.1.101	192.168.1.1	DNS	88	Standard query 0x2780 HTTPS www.google.co.in
18198	52.993726600	192.168.1.101	192.168.1.1	DNS	88	Standard query 0x64d8 A www.google.co.in
18200	52.993836700	192.168.1.101	192.168.1.1	DNS	88	Standard query 0x2bc1 AAAA www.google.co.in
18433	53.513600600	192.168.1.1	192.168.1.101	DNS	115	Standard query response 0x2780 HTTPS www.google.co.in HTTPS
18434	53.513601200	192.168.1.1	192.168.1.101	DNS	106	Standard query response 0x64d8 A www.google.co.in A 142.250.195.67
18435	53.513601600	192.168.1.1	192.168.1.101	DNS	118	Standard query response 0x2bc1 AAAA www.google.co.in AAAA 2404:6800:4000:1017::5e
18439	53.513604200	192.168.1.1	192.168.1.101	DNS	165	Standard query response 0x8cf9 AAAA media.canva.com CNAME media.canva.com.cdn.cloudflare.net AAAA 2606:4700:90d2:883b:1039:4e4:6f
18675	54.050841600	192.168.1.1	192.168.1.101	DNS	179	Standard query response 0x36f1 HTTPS media.canva.com CNAME media.canva.com.cdn.cloudflare.net HTTPS
19515	54.650723800	192.168.1.101	192.168.1.1	DNS	92	Standard query 0x3e1d HTTPS adservice.google.com
19517	54.650817200	192.168.1.101	192.168.1.1	DNS	92	Standard query 0x7ecc AAAA adservice.google.com
19576	54.655585000	192.168.1.101	192.168.1.1	DNS	92	Standard query 0x463c A adservice.google.com
19665	54.685371500	192.168.1.1	192.168.1.101	DNS	119	Standard query response 0x3e1d HTTPS adservice.google.com HTTPS
19666	54.685373400	192.168.1.1	192.168.1.101	DNS	206	Standard query response 0x7ecc AAAA adservice.google.com AAAA 2404:6800:4000:1006::9b AAAA 2404:6800:4000:1006::9c AAAA 2404:6800:4000:1006::9d AAAA 2404:6800:4000:1006::9e
19669	54.685377900	192.168.1.1	192.168.1.101	DNS	158	Standard query response 0x463c A adservice.google.com A 192.178.174.157 A 192.178.174.154 A 192.178.174.155 A 192.178.174.156

○ Observation

- Wireshark first showed **Standard Query** packets.
- Immediately afterward, **Standard Query Response** packets appeared containing the IP addresses.
- Once the browser received the IP address, it began the HTTPS connection.

[Q-5] Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

○ Real-world Scenario (DHCP)

○ Protocol Chosen: DHCP (Dynamic Host Configuration Protocol)

○ Scenario

- When I visit a café and connect my phone to the Wi-Fi network, my phone does not have an IP address.
- The DHCP server automatically assigns:
 1. IP Address
 2. Subnet Mask
 3. Default Gateway
 4. DNS Server

After receiving these network settings, my phone can access the Internet and open apps like Instagram, YouTube, or Spotify without requiring manual network configuration.

This automatic process makes connecting to Wi-Fi fast and easy.